

Reasoning segmentation for videos

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Abstract

This is only a draft of my thesis. I have another document for thesis, where I use document design according to university's rule. Here I only plan and structure my thoughts

1. Introduction

In this chapter, I provide an overview of the motivation, background, and significance of the proposed method for reasoning segmentation for videos, reiterating how it addresses key challenges in a new reason segmentation task for videos. It begins by examining the technical evolution of instance segmentation from convolution neural network to the newest fundamental models. Also, I discuss Multimodal Large Language Models in its current state. The chapter investigates limitations of multimodal LLMs and visionLLMs for the Reasoning segmentation task for videos. Moreover, I talk about current models for Reason Segmentations but for now all these models are used for image segmentation. Finally, it delineates the core objectives and contributions of this thesis, situating the proposed framework VideoReasonSeg.

1.1. Research Background and Significance

To be added

1.2. Related Work

1.2.1. INSTANCE SEGMENTATION MODELS

Here I will talk mostly about RT-DETRv2 [Lv et al. \(2024\)](#) and YOLOv11 [Khanam and Hussain \(2024\)](#)

1.2.2. OPEN-VOCABULARY INSTANCE SEGMENTATION

Not sure for now if I should add separate paragraph for this models. YOLO-world [Cheng et al. \(2024\)](#)

1.2.3. FUNDAMENTAL INSTANCE DETECTION AND SEGMENTATION MODELS

Here I will talk about GroundingDino [Liu et al. \(2024\)](#) and Segment anything model 2 [Ravi et al. \(2024\)](#)

1.2.4. MULTIMODAL LLMs

Right now the most advanced should be QWen 2.5 [Bai et al. \(2025\)](#) video LLAMA 3 [Zhang et al. \(2025\)](#)

1.2.5. REASONING SEGMENTATION FOR IMAGES

Current models are LISA [Lai et al. \(2024\)](#) and GSVA [Xia et al. \(2024\)](#)

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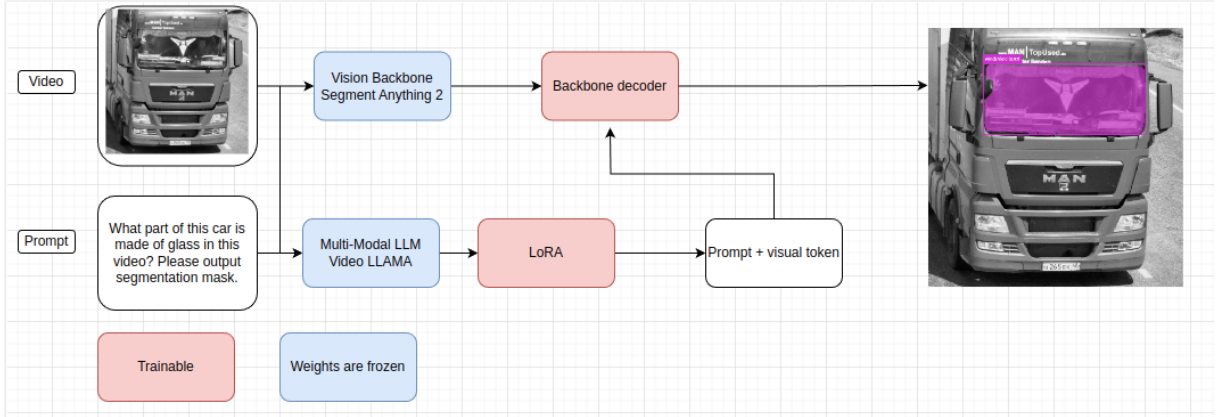


Figure 1: Framework pipeline example

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